Numerische Mathematik II für Ingenieurwissenschaften

Lecture 1 – Introduction and organizational things

Prof. Dr. Tobias Breiten, Prof. Dr. Sandra May

Institute of Mathematics, TU Berlin

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Overview

1. Applications
2. The team
3. Class outline

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Why numerical methods for PDEs?

(Flow) phenomena in nature

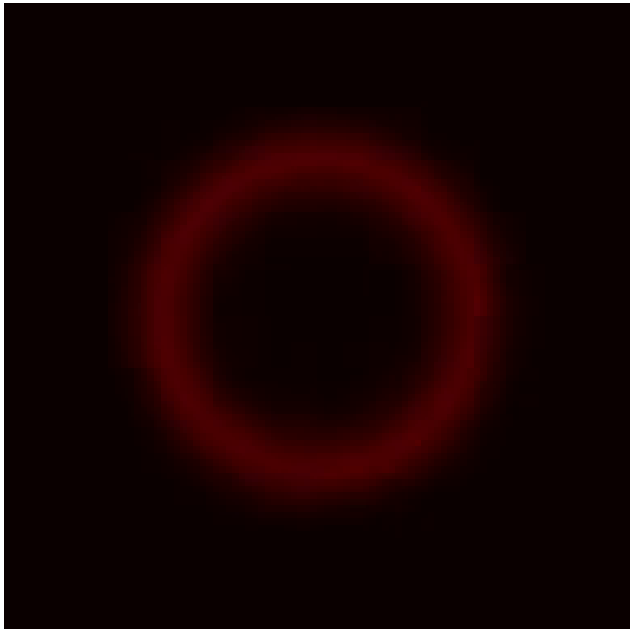
$\Rightarrow$ *Physicist / Mathematician:*
Describe them using partial differential equations (PDEs)

Issue: Too complicated for solving them with pen and paper
$\Rightarrow$ **solve** them **approximately** on the computer...

Heat equation

Consider temperature profile $v(t,x)$ of a heated plate described by the heat equation

$$
\begin{aligned}
\tfrac{\partial v}{\partial t}(t,x) &= \Delta v(t,x) + f_{\mathrm{burn}}(t,x), && \text{in } (0,\infty)\times\Omega,\\
v(0,x) &= 0, && \text{in } \Omega,\\
v(t,x) &= 0, && \text{on } \partial\Omega.
\end{aligned}
$$



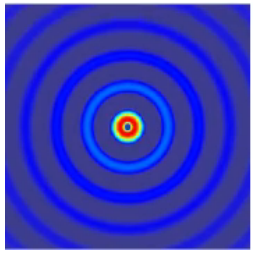
Stove top burner[1]

[1] By Cale.rankin - Own work, CC BY-SA 4.0, https://commons.wikimedia.org/w/index.php?curid=67833967

Wave equation

Consider displacement $u(t,x)$ from rest situation described by the wave equation

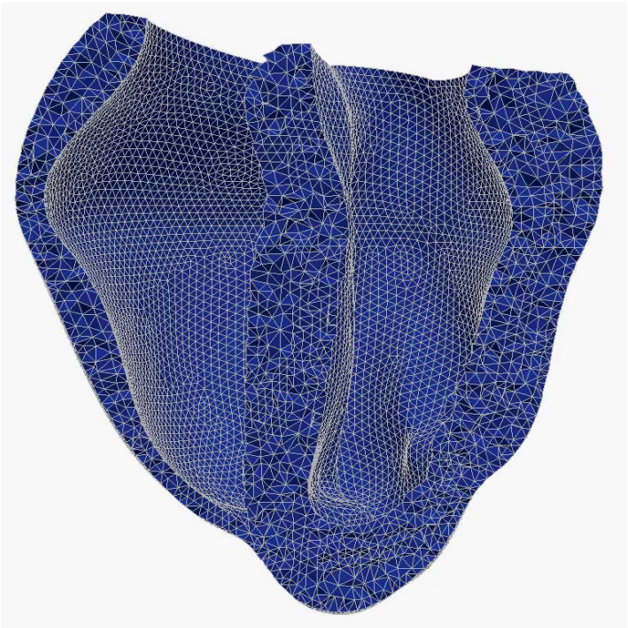
$$\begin{aligned} \frac{\partial^2 u}{\partial t^2}(t,x) &= c^2 \Delta u(t,x) + f_{\mathrm{src}}(t,x), && \text{in } (0,\infty) \times \mathbb{R}^2, \\ u(0,x) &= 0, && \text{in } \mathbb{R}^2. \end{aligned}$$



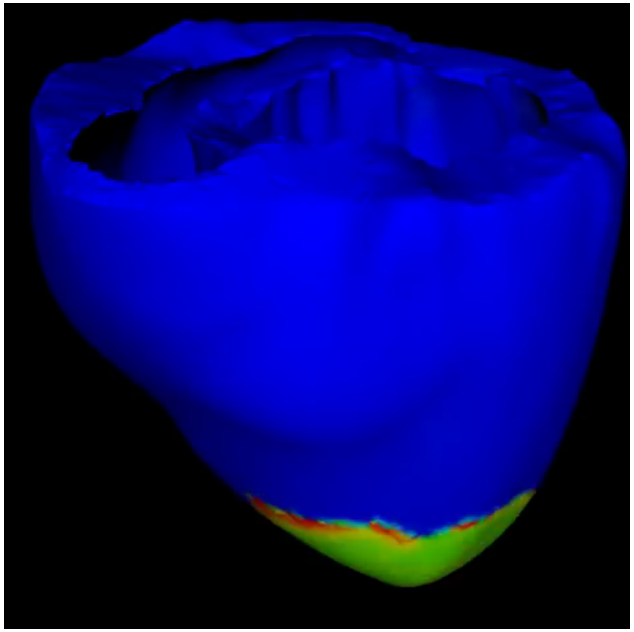
Spherical waves from point source[2]

[2] By Oleg Alexandrov - self-made with MATLAB, Public Domain, https://commons.wikimedia.org/w/index.php?curid=3036844

Bidomain equations ("real life" version)



Regular heart beat[3]



Arrhythmia[4]

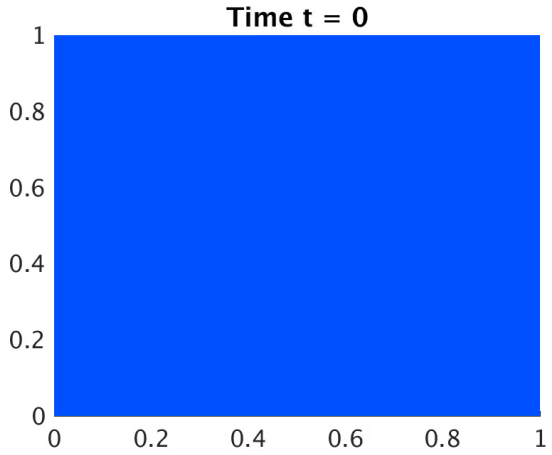
[4] Chaste Project. (2011/03/30). *Adaptive heart simulation*. YouTube.

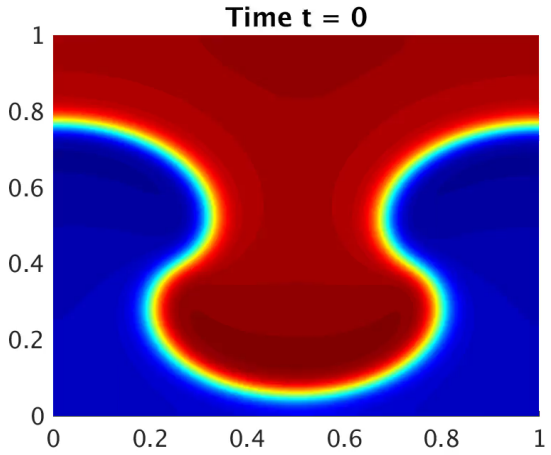
[5] cardiosolv. (2011/02/01). *Fibrillation with Electrical and Mechanical Activity in the Canine Ventricles*. YouTube.

Bidomain equations ("mathematicians life" version)

Model for the electric potential in the human heart

$$
\begin{aligned}
\frac{\partial v}{\partial t} &= \operatorname{div}(M_i \nabla v_i) + I_{\mathrm{ion}}(v,w) + I_i, && \text{in } (0,\infty)\times\Omega,\\
\frac{\partial v}{\partial t} &= -\operatorname{div}(M_e \nabla v_e) + I_{\mathrm{ion}}(v,w) - I_e, && \text{in } (0,\infty)\times\Omega,\\
\frac{\partial w}{\partial t} &= G(v,w), && \text{in } (0,\infty)\times\Omega,\\
\nu\cdot M_i\nabla v_i &= 0, \quad \nu\cdot M_e\nabla v_e = 0, && \text{on } \partial\Omega\times(0,\infty),\\
v(x,0) &= v_0(x), \quad w(x,0) = w_0(x), && \text{in } \Omega.
\end{aligned}
$$



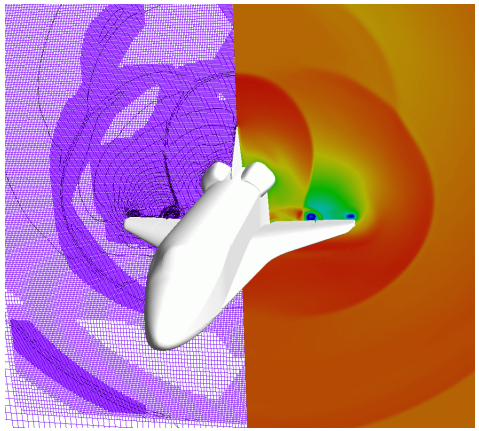


High speed gas flow

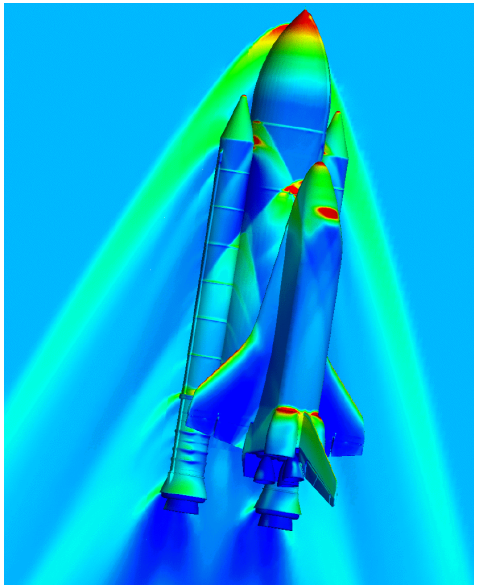
Approach: Model using the compressible Euler equations

In 1d:

$$\begin{bmatrix} \varrho \\ \varrho u \\ E \end{bmatrix}_t + \begin{bmatrix} \varrho u \\ \varrho u^2 + p \\ (E+p)u \end{bmatrix}_x = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$



(Berger,Aftosmis)



(Berger,Aftosmis)

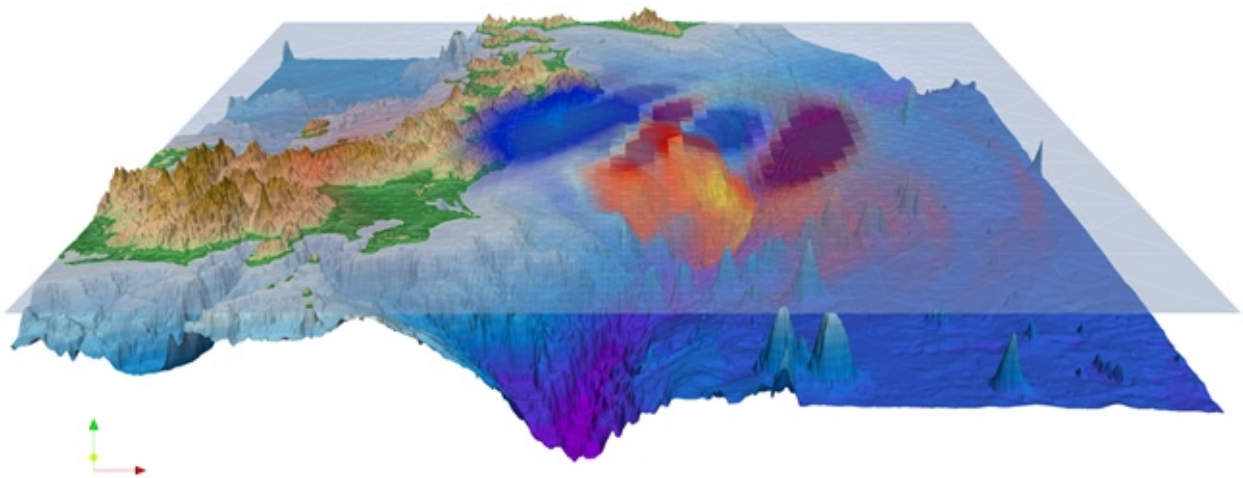
Simulation of tsunami

Approach: Model using the shallow water equations

$$\frac{\partial(\varrho\eta)}{\partial t} + \frac{\partial(\varrho\eta u)}{\partial x} + \frac{\partial(\varrho\eta v)}{\partial y} = 0,$$

$$\frac{\partial(\varrho\eta u)}{\partial t} + \frac{\partial}{\partial x}\left(\varrho\eta u^2 + \frac{1}{2}\varrho g\eta^2\right) + \frac{\partial(\varrho\eta uv)}{\partial y} = 0,$$

$$\frac{\partial(\varrho\eta v)}{\partial t} + \frac{\partial(\varrho\eta uv)}{\partial x} + \frac{\partial}{\partial y}\left(\varrho\eta v^2 + \frac{1}{2}\varrho g\eta^2\right) = 0.$$

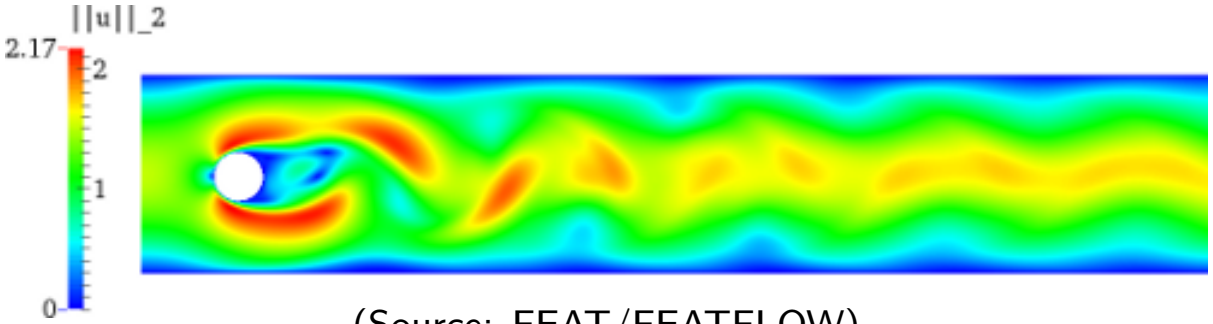


(siam news)

Flow problem with viscous effects

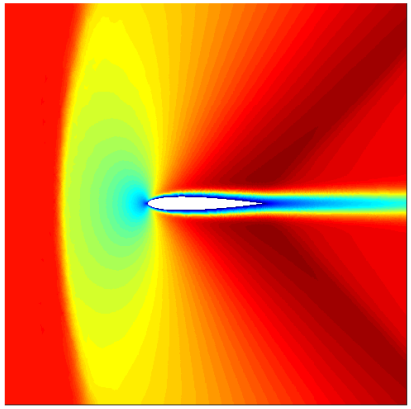
Equations: Navier-Stokes equations

Incompressible:



(Source: FEAT/FEATFLOW)

Compressible:



Cool flow movies

using the software package `deal.II`: by Matthias Maier

`https://www.youtube.com/@matthiasmaier8956/featured`

Mathematical modelling and simulation of flow problems

Depending on

- how strong the viscous effects are,
- how strong the convection is,
- ...

you need different / appropriate mathematical models

and different numerical methods to solve them

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Depending on

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and different numerical methods to solve them

Each numerical method has its strength and weaknesses!

In this course: introduction to several wide-spread numerical methods.

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Prof. Dr. Tobias Breiten

Professor for *Modeling, Simulation and Optimization of Real Processes*

- Diplom in Technomathematik, TU Kaiserslautern
- PhD in Mathematics, OvGU Magdeburg
- Habilitation in Mathematics, KFU Graz
- Since 04/2020: Professor, TU Berlin

Prof. Dr. Sandra May

Professor for *Numerical Methods for PDEs*

- Diplom in Mathematics, Heidelberg
- PhD in Mathematics, Courant Institute, NYU
- Postdoc, ETH Zurich
- Juniorprofessor, TU Dortmund University
- Associate Professor, Uppsala University
- Since 09/2024: Professor, TU Berlin

Dr. Juliette Dubois

Postdoc in group *Numerical Methods for PDEs*

- M.Sc. in Mathematics in Paris, France and Vienna, Austria
- PhD in Mathematics in Paris, France
- Postdoc, RWTH Aachen
- Since 03/2025: Postdoc, TU Berlin

Alireza Selahi Moghaddam

Teaching Assistant

- B.Sc. + M.Sc. in *Physikalische Ing.wissenschaften*
- B.Sc. in Mathematics
- Ongoing: M.Sc. in Mathematics

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Three prototypes of second-order PDEs

- Poisson equation: elliptic

$$-\Delta u = f \quad \text{in } \Omega$$

+ suitable boundary conditions

- Heat equation: parabolic

$$\partial_t u - \Delta u = f \quad \text{in } \Omega$$

+ initial conditions and suitable boundary conditions

- Wave equation: hyperbolic

$$\partial_{tt} u - c^2 \Delta u = f \quad \text{in } \Omega$$

+ initial conditions and suitable boundary conditions

Numerical methods for PDEs

- Many many methods out there...
- Some of the most wide-spread and most general methods:
  - Finite Difference (FD) methods
  - Finite Volume (FV) methods
  - Finite Element (FE)/ Galerkin methods

Numerical methods for PDEs

- Many many methods out there...
- Some of the most wide-spread and most general methods:
  - Finite Difference (FD) methods
  - Finite Volume (FV) methods
  - Finite Element (FE)/ Galerkin methods
- Plan for this course
  - Notation, classification and basic theory of PDEs
  - Introduction to FD
  - Introduction to FV
  - Introduction to continuous FE

(Rough plan of) Course structure

Part I: taught by Sandra May

- Roughly 4-5 lectures
- FD schemes for different model problems

Part II: taught by Tobias Breiten

- Roughly until Xmas
- Notation, classification and basic theory of PDEs
- Focus method: FE schemes
- Focus PDE: elliptic (and parabolic) PDEs

Part III: taught by Sandra May

- Remaining time
- Focus method: FV schemes
- Focus PDE: (first-order) hyperbolic conservation laws

Lecture and exercise class

Lectures:

- Tue, 8-10, MA041
- Th, 8-10, MA042
- When S.May teaches: start at 08:20 (!)

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Lecture Notes:

- Currently: Restructuring of course
- We are working on typed lecture notes; hope to have draft by end of course

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Exercise classes:

- Lead by Juliette Dubois and Alireza Selahi Moghaddam
- Time of office hour by Juliette and Alireza will be announced
- Precise time to be determined (information follow now)

Registration for exercise class

Deadline 1: Registration to the tutorial before **15.10.** until 18:00

- The tutorial schedule will be available on 17.10
- *LiveDemo: Go into ISIS and MOSES now!*

Deadline 2: Enter your homework group in Isis before **24.10.**

Homework and exercise class

Homework:

- New homework problem sheet every week on Friday
- Will involve programming problems using Python
- Do homework problems in groups of 3
- Hand in on ISIS: deadline on Sunday, group feedback the week after
- Problem Sheet 0 (not to be handed in): due in week of Oct 20-24

Exercise class:

- Mainly: discusses previous homework
- Pre-discuss new homework, review material / chance to ask questions
- Start of exercise classes: week of Oct 20-24
- Before first exercise: install Python, NumPy and Matplotlib!

Written exam

Exam:

- 90 minutes
- No exam aids allowed
- Written (paper) exam
  BUT will contain programming / practical exercises, for example
  - write pseudo code
  - find mistakes in given codes
  - interpret numerical results / identify correct ones
  - ...
    (We are aware that you do not have compiler.)

Exam dates:

- Exam 1: Tue, 24.02.2026, 14:00 – 15:30 (registration starts October 27)
- Exam 2: Wed, 25.03.2026, 10:00 – 11:30 (registration start early March)
  (arrive earlier – more information later)

Communication

- Communication and information are on the course homepage in ISIS
- *Live Demo: Go into ISIS*
  - Announcement forum
  - Discussion forum
  - Homework group forum

Questions?

Mentimeter

- Option 1: Go on `www.menti.com` and enter **8872 7780**
- Option 2: Scan